

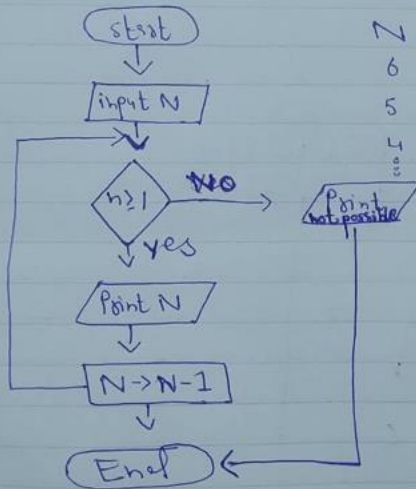
Assignment #1

Programming Fundamentals:

Q#1: Algorithm:

- ① Start
- ② Take input N
- ③ Print N
- ④ $N \rightarrow (n-1)$ loop
- ⑤ Repeat step 3 and 4 until $N=0$
- ⑥ End

Flowchart:



Dry Run:

N	n → n-1	output
6	6-1	5
5	5-1	4
4	4-1	3
...	...	...

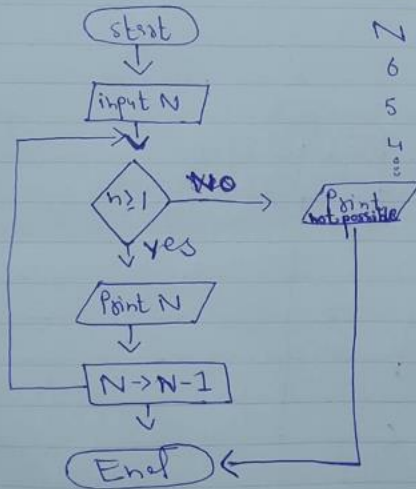
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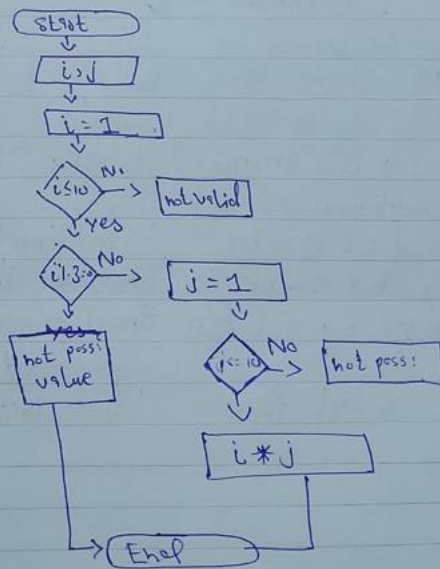
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4	4-1	3
...	...	...

Q #3: Algorithm:

- ① start
- ② take input i and j
- ③ First we look variable i and satisfy the condition:
if $i \leq 10$, so take process onwards otherwise
Print this is invalid number.
- ④ Now we will take variable j and satisfy condition
if $j \leq 10$ else Print condition not met.
- ⑤ Now we multiply both numbers and satisfied the
conditions.
- ⑥ Print $i * j$
- ⑦ End

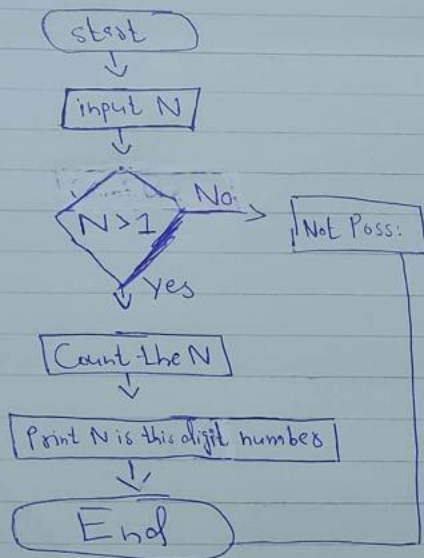
=> Flowchart:



Q#4: Algorithm:

- ① Start
- ② Take input N
- ③ Count the number of digit
- ④ if $N > 1$; 2, 3, 4, this satisfy the condition.
- ⑤ Print it is this digit no.
- ⑥ Else Print not possible
- ⑦ End.

Flowchart:



Q#5: Algorithm:

- ① Start
- ② Take input N
- ③ Now we will set $sum = 0$ and $main = N$
- ④ if $main > 0$
- ⑤ $digit = main \% 10$, $sum = sum + digit$, $main = main / 10$
- ⑥ if ~~Yes~~ then take input set $N = sum$ if $n \geq 10$ & if yes ~~breaks~~ the condition - set $N = sum$
- ⑦ else print the final single digit
- ⑧ End.

Flowchart:

